

GRADE 11 • ALGEBRA II • TOPIC 6

Lesson 6-3

Logarithms

Exponential and Logarithmic Functions

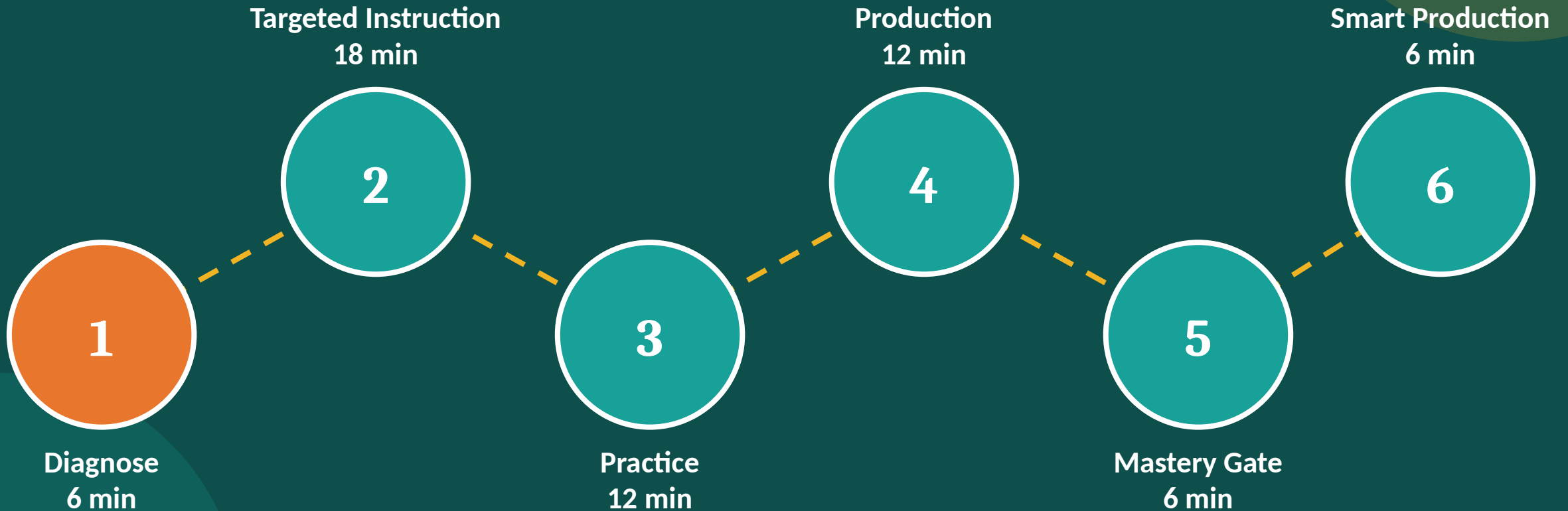
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$$\log_b x = y \Leftrightarrow b^y = x$$

Journey of 6 stations, 60 minutes

Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Understand and evaluate logarithms.
2. Analyze common logarithms and natural logarithms.
3. Convert between exponential and logarithmic forms.
4. Use logarithms to solve problems involving exponential models.
5. Evaluate logarithms using technology.

VOCABULARY

common logarithm

logarithm

logarithmic function

natural logarithm

order of magnitude

STANDARDS

HSF.BF.B.4.A Solve an equation of the form $f(x) = c$ for a simple function f that has an inverse and write an expression for the inverse.

HSF.BF.B.5 Understand the inverse relationship between exponents and logarithms and use this relationship to solve problems involving logarithms and exponents.

HSF.LE.A.4 For exponential models, express as a logarithm the solution to $ab^{(ct)} = d$ where a , c , and d are numbers and the base b is 2, 10, or e ; evaluate the logarithm using technology.

ESSENTIAL QUESTION

How can we use logarithms and relate them to exponents?

MATHEMATICAL PRACTICES MP.2 · MP.5

STATION 1 • DIAGNOSE • 6 MIN

Start here: three quick checks

1

Evaluate 2^5

2

Solve $3^x = 81$

3

$2^x = 20$: x is between which integers?

Done when

Done when: all three are answered without a calculator, and for the last one you can name the two powers of 2 that trap the answer.

STATION 2 • TARGETED INSTRUCTION • 18 MIN

A Logarithm Is an Exponent

$$\log_b x = y \iff b^y = x \quad b > 0, b \neq 1, x > 0$$

EXAMPLE A

Write $3^4 = 81$ as a logarithm
base 3, exponent 4, result 81

$$\log_3 81 = 4$$

EXAMPLE B

Evaluate $\log_2 32$

$$2^y = 32$$

$$2^5 = 32$$

$$\log_2 32 = 5$$

EXAMPLE C

Evaluate $\log_7 \frac{1}{49}$

$$7^y = \frac{1}{49}$$

$$7^{-2} = \frac{1}{49}$$

$$\log_7 \frac{1}{49} = -2$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Common and Natural Logarithms

$$\log x = \log_{10} x \quad \ln x = \log_e x, \quad e \approx 2.718$$

EXAMPLE A

Evaluate $\log 1000$

$$10^y = 1000$$

$$\log 1000 = 3$$

EXAMPLE B

Evaluate $\log 0.001$

$$10^y = \frac{1}{1000} = 10^{-3}$$

$$\log 0.001 = -3$$

EXAMPLE C

Evaluate $\ln e^5$

$$e^y = e^5$$

$$\ln e^5 = 5$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Evaluating with Technology

integer powers: convert and read off otherwise: use LOG, LN or the log key

EXAMPLE A

Evaluate $\log 50$ and $\ln 50$

$$\log 50 \approx 1.699$$

$$\ln 50 \approx 3.912$$

EXAMPLE B

Estimate, then
evaluate $\log_2 20$

$$2^4 = 16 < 20 < 2^5 = 32$$

$$4 < \log_2 20 < 5$$

$$\log_2 20 \approx 4.322$$

EXAMPLE C

Order of magnitude
of 6.3×10^8

$$\log(6.3 \times 10^8) \approx 8.80$$

rounds to 9 $\Rightarrow 10^9$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Logarithms and Exponential Models

$$b^t = k \iff t = \log_b k \quad A = A_0 b^t \Rightarrow t = \log_b \frac{A}{A_0}$$

EXAMPLE A

Solve $5 \cdot 2^t = 40$

$$2^t = 8$$

$$t = \log_2 8 = 3$$

EXAMPLE B

Solve $1000(1.05)^t = 2000$

$$1.05^t = 2$$

$$t = \log_{1.05} 2$$

$$t \approx 14.2$$

EXAMPLE C

Solve $200e^{0.03t} = 600$

$$e^{0.03t} = 3 \Rightarrow 0.03t = \ln 3$$

$$t = \frac{\ln 3}{0.03}$$

$$t \approx 36.6$$

STATION 3 • PRACTICE • 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Write $4^3 = 64$ as a logarithm

Evaluate $\log_2 \frac{1}{8}$

Done when: each answer is checked by rewriting it in exponential form.

APPLY

Use it in context

$$6000(1.18)^t = 50000$$

Solve for t (1 d.p.)

Done when: the equation is set up before using a calculator, and the answer states units (years).

INVESTIGATE

Find out why

Why is $\log_b x$ undefined for $x \leq 0$?

Show that $\log_b(b^k) = k$

Done when: your argument uses what a logarithm means, not only that the calculator gives an error.

STATION 4 • PRODUCTION • 12 MIN

Production: Red Sea Visitors: When Do We Reach 3 Million?

THE MODEL

A Red Sea destination under Vision 2030 plans for growth in yearly visitors. In this model, the destination welcomed 1.2 million visitors in the base year 2026 and visitor numbers grow by 12 percent every year. All numbers are model values, not official statistics. Time t is measured in years after 2026.

$$V(t) = 1.2(1.12)^t \text{ (millions)}$$

$$V(t) = 3$$

1. Write the equation to be solved for t and isolate the exponential term.
2. Use a logarithm to write the exact solution, then evaluate it to one decimal place.
3. Check by substitution, state the calendar year, and explain why dividing 3 by 1.2 does not answer the question.

DONE WHEN

- The exponential term is isolated before any logarithm is taken.
- The exact logarithm answer is written before the decimal.
- The answer is checked by substituting back into the model.
- The conclusion names units and the year, and says it is a model value.

Hint for solving: divide both sides by 1.2 first. Then ask: 1.12 to what power gives 2.5?

STATION 5 • MASTERY GATE • 6 MIN

Mastery Gate: three questions to pass

1

Evaluate $\log_4 \frac{1}{16}$

2

Evaluate $\ln e^3 + \log 100$

3

Solve $500(1.04)^t = 1500$ (1 d.p.)

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 • SMART PRODUCTION • 6 MIN

Smart Production: make one, then check it

CREATE

1. Design a one-slide model for a Saudi growth or decay situation of your own (label every number as a model value).
2. Show the equation, the isolated exponential, the exact logarithm answer and the decimal with units.
3. Add one NON-example: dividing the two values and stopping, with a caption saying what that shortcut actually computes.

Done when: your slide is checked by substitution and a partner can follow it without you.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Solve $120e^{0.05t} = 360$ exactly
Give t to 1 d.p.

Next lesson: 6-4 Logarithmic Functions.

Exam connections

SAT

$$500(1.06)^t = 1000 \Rightarrow t = ?$$

STEPS

1 $1.06^t = 2$

2 $t = \log_{1.06} 2 \approx 11.9$

GAT

Compare $\log_2 64$ and $\log_3 243$

STEPS

1 $2^6 = 64 \Rightarrow \log_2 64 = 6$

2 $3^5 = 243 \Rightarrow \log_3 243 = 5$

3 $6 > 5$: first is greater

SAAT

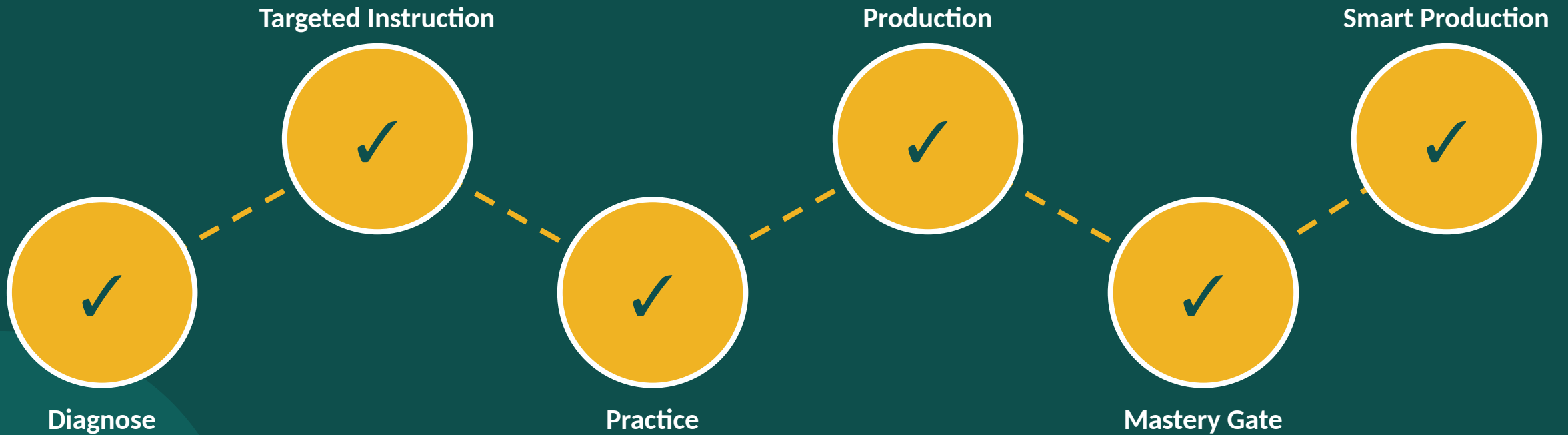
If $\log_4 x = 3$, find $\log_2 x$

STEPS

1 $x = 4^3 = 64$

2 $\log_2 64 = 6$

Journey complete



A logarithm is an exponent: convert, evaluate with LOG or LN, and use it to find time in exponential models.